

CV Maker — Tech Stack and Dependencies

Project: CV Maker
Technology: Python/FastAPI backend, React/Vite + Svelte frontends, PostgreSQL, SQLAlchemy, Playwright scraper
Location: /home/dominic/Documents/CV MAKER
Report #: 5/20

Primary Stack

Python/FastAPI backend, React/Vite + Svelte frontends, PostgreSQL, SQLAlchemy, Playwright scraper

Backend Dependencies

- fastapi — API framework
- sqlalchemy — ORM
- psycopg2 — PostgreSQL driver
- playwright — web scraper
- jinja2 — PDF templating
- weasyprint — PDF generation
- python-jose — JWT
- passlib — password hashing

Frontend (React): react, vite, react-router-dom, axios, tailwindcss

Frontend (Svelte): svelte, vite, svelte-routing, chart.js

Technology Stack Details

Framework Versions

Component	Technology	Version	Purpose
Backend	FastAPI	API server	
Database	PostgreSQL	Persistence	
Auth	JWT	Security	

Component	Technology	Version	Purpose
Frontend	React	UI	
Hosting	Render.com + GitHub Pages	Deployment	

Core Dependencies

```
# Core dependencies
fastapi>=0.115.0
sqlalchemy>=2.0.0
pydantic>=2.0.0
uvicorn[standard]>=0.30.0
```

Dependency Justification

Package	Why	Alternative Considered
fastapi	Modern ASGI framework	Flask, Django
sqlalchemy	Mature ORM with async	peewee, tortoise-orm
pydantic	Validation + serialization	marshmallow
jose	JWT handling	PyJWT

Development Dependencies

Package	Purpose
pytest	Testing framework
httpx	Async HTTP client for tests
uvicorn	ASGI server
python-dotenv	Environment variable loading

External Services

Service	Purpose	Integration Point
OpenAI	AI chat completion	/v1/chat/completions proxy
GitHub Pages	Static hosting	git push deployment

Version Constraints

- Python 3.10+ required (type hints, pattern matching)
- Pydantic v2 for improved validation performance
- SQLAlchemy 2.0+ for modern async support